

# Kushin Mukherjee

Postdoctoral scholar

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## Appointments

2025— *Postdoctoral scholar*, Psychology, Stanford University  
Advisor: Judith E. Fan



## Education

2019-2025 *PhD*, Psychology, University of Wisconsin-Madison  
Advisors: Timothy T. Rogers, Karen B. Schloss



2015-2019 *AB*, Cognitive Science and Japanese, minor in Mathematics, Vassar College  
Thesis advisor: Joshua R. de Leeuw  
*general honors*  
*departmental honors in Cognitive Science and Japanese*



## Grants, honors, & awards

2025 Best Paper Honorable Mention, **IEEE VIS**  
2025 Computational Modeling Prize in Applied Cognition, **Cognitive Science Society**  
2025 Best Paper Award, **ICLR Workshop on Bidirectional Human-AI Alignment**  
2025 Student Research Grants Competition Travel Award, **UW-Madison**  
2025 Distinguished Paper Award, McPherson Eye Research Institute, **UW-Madison**  
2021-2025 Hertz Travel Award, Department of Psychology, **UW-Madison**  
2021 Center for Brain, Minds, and Machines Summer School Fellow, **MIT**  
2021 Best Paper Honorable Mention, **IEEE VIS**  
2021 Kenzi Valentyn Vision Research Award, McPherson Eye Research Institute, **UW-Madison**  
2021 Elsevier Travel Award, **Vision Sciences Society**  
2020-2022 Marie Christine Kohler Fellow, Wisconsin Institute for Discovery, **UW-Madison**  
2019 Yin-Lien C. Chin Prize for best senior project in Chinese or Japanese, **Vassar College**  
2019 Phi Beta Kappa, **Vassar College**  
2019 Sigma Xi, **Vassar College**  
2018 CSLI Summer Intern, **Stanford University**  
2018 Psi Chi, **Vassar College**  
2016 Summer Program Scholarship, **Ochanomizu University**  
2016 Japan Student Service Organization Scholarship

2015-2019 Sarah Tod Fitz Randolph Scholarship Fund, Vassar College

## Research Experience

06 - 09, 2024 **AI/ML Intern**, Apple  
08, 2021 **Summer School Fellow**, MIT Center for Minds, Brains, and Machines  
06 - 08, 2018 **CSLI Summer Intern**, Stanford University

## Working Papers & Preprints

- Studdiford, Z., Rogers, T. T., Suresh, S.\*, & **Mukherjee, K.\*** (*under review*). Evaluating Steering Techniques using Human Similarity Judgments.
- Studdiford, Z., Rogers, T. T., **Mukherjee, K.\***, & Suresh, S.\* (*under review*). Uncovering the Computational Ingredients of Human-Like Representations in LLMs.
- Chen, A., Kim, S. S. Y., Franyutti, A., Dharmasiri, A., **Mukherjee, K.**, Fan, J. E., & Russakovsky, O. (*under review*). Presenting large language models as companions affects what mental capacities people attribute to them.
- **Mukherjee, K.**, Rogers, T. T., Lessard, L., Gleicher, M., & Schloss, K. B. (*under review*). Color-concept associations reveal a new conceptual space.
- **Mukherjee, K.**, Huey, H., Hebart, M. N., Fan, J. E., & Bainbridge, W. A. (*under review*). Drawings-of-THINGS: A large-scale dataset containing human sketches of 1,854 object concepts.
- **Mukherjee, K.**, Rogers, T. T., & Schloss, K. B. (*under revision*). Estimating human color-concept associations from multimodal language models.
- Verma, A., **Mukherjee, K.**, Potts, C., Kreiss, E., & Fan, J. (*under revision*). CHART-6: Human-Centered Evaluation of Data Visualization Understanding in Vision-Language Models.
- **Mukherjee, K.**, Yin, B., Lessard, L., & Schloss, K. B. (*in prep*). How do people map colors to concepts? Modeling assignment inference as evidence accumulation.

## Peer-reviewed Publications

- 2025 Braun, H., **Mukherjee, K.**, Gorelik, S. R. & Schloss, K. B. (2025). Affective color scales for colormap data visualizations. *IEEE Transactions on Visualization and Computer Graphics*. **\*Best paper honorable mention award**
- 2025 **Mukherjee, K.**, Ren, D., Moritz, D. & Assogba, Y. (2025). EncQA: Evaluating visual encoding understanding for visualizations in large vision-language models. *IEEE Transactions on Visualization and Computer Graphics*.
- 2025 **Mukherjee, K.**, & Rogers, T. T. (2025). Using drawings and deep neural networks to characterize the building blocks of human visual similarity. *Memory & Cognition*.

- 2023 Suresh, S., **Mukherjee, K.**, Yu, X., Huang, W., Padua, L., & Rogers, T. T. (2023). Conceptual structure coheres in human cognition but not in large language models. *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- 2023 **Mukherjee, K.**, Lu, X., Huey, H., Vinker, Y., Shamir, A., & Fan, J. E. (2023). SEVA: Leveraging sketches to evaluate alignment between human and machine visual abstraction. *Advances in Neural Information Processing Systems (NeurIPS), Datasets & Benchmarks Track*.
- 2021 **Mukherjee, K.**, Yin, B., Sherman B. E., Lessard, L. & Schloss, K. B. (2021). Context matters: Semantic discriminability theory for perceptual encoding systems. *IEEE Transactions on Visualization and Computer Graphics*. **\*Best paper honorable mention award**

## Peer-reviewed Proceedings & Workshop Papers

- 2025 Suresh, S.\*, **Mukherjee, K.\***, Giallanza, T., Yu, X., Patil, M., Cohen, J. D., & Rogers, T. T. (2025). AI-assisted semantic norms for 786 concepts. *ICLR Workshop on Bidirectional Human-AI Alignment*. **\*Best paper award**
- 2025 Suresh, S.\*, **Mukherjee, K.\***, Giallanza, T., Yu, X., Patil, M., Cohen, J. D., & Rogers, T. T. (2025). AI-assisted semantic norms for 786 concepts. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*. **\*Modeling prize in applied cognition**
- 2024 **Mukherjee, K.** (2024). Shaping vision through drawing. *Nature Reviews Psychology*.
- 2024 Verma, A., **Mukherjee, K.**, Kreiss, E., Potts, C., and Fan, J. (2024). Evaluating human and machine understanding of data visualizations. *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*.
- 2024 **Mukherjee, K.**, Suresh, S., Yu, X., & Lupyan, G. (2024). The role of shared labels and shared experiences in representational alignment. *International Conference on Learning Representations (ICLR) Re-Align Workshop*
- 2024 Suresh, S., Huang, W., **Mukherjee, K.**, & Rogers, T.T. (2024). Categories vs semantic features: What shapes the similarities people discern in photographs of objects? *International Conference on Learning Representations (ICLR) Re-Align Workshop*
- 2023 **Mukherjee, K.**, Kim, N. Y., Alamooti, S. T., Adolphs, R., & Kar, K. (2023). Leveraging Artificial Neural Networks to Enhance Diagnostic Efficiency in Autism Spectrum Disorder: A Study on Facial Emotion Recognition. *Conference on Cognitive Computational Neuroscience*.
- 2023 **Mukherjee, K.**, Lu, X., Huey, H., Vinker, Y., Shamir, A., & Fan, J. E. (2023). Evaluating machine comprehension of sketch meaning at different levels of abstraction. *Proceedings of the 45th Annual Meeting of the Cognitive Science Society*.
- 2023 Suresh, S., **Mukherjee, K.** & Rogers, T. T. (2023). Semantic Feature Verification in FLAN-T5. *International Conference on Learning Representations (ICLR), Tiny Papers Track*.
- 2023 **Mukherjee, K.**, Suresh, S. & Rogers, T. T. (2023). Human-machine cooperation for semantic feature generation. *International Conference on Learning Representations (ICLR), Tiny Papers Track*.

- 2020 **Mukherjee, K.**, & Rogers, T. T. (2020). How does task structure shape representations in deep neural networks? *2nd NeurIPS Workshop on Shared Visual Representations in Human and Machine Intelligence*.
- 2019 **Mukherjee, K.**, Hawkins, R. D., & Fan, J. E. (2019). Communicating semantic part information in drawings. *Proceedings of the 41st Annual Meeting of the Cognitive Science Society*.

## Book Chapters

- 2024 Schloss, K. B., Schoenlein, M. A., & **Mukherjee, K.** (2024). Color semantics for visual communication. In R. B. D. A. Szafr, M. Chen, D. J. Edward, B. Fisher, & L. M. Padilla (Ed.), *Visualization Psychology*.

## Conference Presentations

- 2025 **Mukherjee, K.**, Ren, D., Moritz, D. & Assogba, Y. (2025). EncQA: Evaluating visual encoding understanding for visualizations in large vision-language models. Talk presented at IEEE VIS 2025.
- 2025 Braun, H., **Mukherjee, K.**, Gorelik, S. R. & Schloss, K. B. (2025). Affective color scales for colormap data visualizations. Talk presented at IEEE VIS 2025.
- 2025 Chinni, A., Colón, Y. I., **Mukherjee, K.**, Gleicher, M. L., & Schloss, K. B. (2025). Development of the NBT-53 Texture Library for the Study of Texture Semantics. Poster presented at the 66th Annual Meeting of the Psychonomic Society.
- 2025 Braun, H., **Mukherjee, K.**, Gorelik, S. R. & Schloss, K. B. (2025). Affective color scales for colormap data visualizations. Poster presented at the 25th Annual Meeting of the Vision Sciences Society.
- 2025 Chinni, A., Colón, Y. I., **Mukherjee, K.**, Gleicher, M. L., & Schloss, K. B. (2025). Development of the NBT-53 Texture Library for the Study of Texture Semantics. Poster presented at the 25th Annual Meeting of the Vision Sciences Society.
- 2025 Suresh, S. **Mukherjee, K.**, Giallanza, T., Yu, X., Patil, M., Cohen, J. D., & Rogers, T. T. (2025). AI-assisted semantic norms for 786 concepts. Talk presented at the 47th Annual Meeting of the Cognitive Science Society.
- 2025 Li, Z., Colón, Y. I., **Mukherjee, K.**, Rogers, T. T. (2025). Iterated LASSO reveals highly distributed and variable representations of faces, places, and objects. Poster presented at the 47th Annual Meeting of the Cognitive Science Society.
- 2025 Suresh, S. **Mukherjee, K.**, Giallanza, T., Yu, X., Patil, M., Cohen, J. D., & Rogers, T. T. (2025). AI-assisted semantic norms for 786 concepts. Talk and Poster presented at the Bi-Align Workshop at ICLR.
- 2024 **Mukherjee, K.**, Rogers, T. T., & Schloss, K. B. (2024). Estimating human color-concept associations from multimodal language models. Poster presented at the 46th Annual Meeting of the Cognitive Science Society.
- 2024 Verma, A., **Mukherjee, K.**, Kreiss, E., Potts, C., and Fan, J. (2024). Evaluating human and machine understanding of data visualizations. Poster presented at the 46th Annual Meeting of the Cognitive Science Society.

- 2024 Suresh, S., **Mukherjee, K.**, & Rogers, T. T. (2024). Can deep convolutional networks explain the semantic structure that humans see in photographs? Poster presented at the 46th Annual Meeting of the Cognitive Science Society.
- 2023 **Mukherjee, K.**, Kim, N. Y., Alamooti, S. T., Adolphs, R., & Kar, K.. (2023). Leveraging Artificial Neural Networks to Enhance Diagnostic Efficiency in Autism Spectrum Disorder: A Study on Facial Emotion Recognition. Talk and Poster presented at the Conference on Cognitive Computational Neuroscience.
- 2023 **Mukherjee, K.**, Lessard, L., & Schloss K. B. (2023). How do people map colors to concepts? Modeling assignment inference as evidence accumulation. Talk presented at the 23rd Annual Meeting of the Vision Sciences Society.
- 2023 Suresh, S., **Mukherjee, K.**, & Rogers T. T. (2023). Can deep convolutional networks explain the semantic structure that humans see in photographs? Talk presented at the 23rd Annual Meeting of the Vision Sciences Society.
- 2023 Fan, J. E., **Mukherjee, K.**, Huey, H., Hebart, M., & Bainbridge, W. (2023). THINGS-drawings: A large-scale dataset containing human sketches of 1,854 object concepts. Talk presented at the 23rd Annual Meeting of the Vision Sciences Society.
- 2023 **Mukherjee, K.**, Lu, X., Huey, H., Vinker, Y., Shamir, A., & Fan, J. E. (2023). Evaluating machine comprehension of sketch meaning at different levels of abstraction. Poster presented at the 23rd Annual Meeting of the Vision Sciences Society.
- 2022 Armendariz, M., **Mukherjee, K.**, Shang, J., & Kar, K. (2022). Probing the functional relevance of side-reads and bypass-connections in the primate ventral stream during visual object recognition using deep neural networks. Poster presented at the 22nd Annual Meeting of the Vision Sciences Society.
- 2022 **Mukherjee, K.**, Schloss, K. B., Lessard, L., Gleicher, M., & Rogers, T.T. (2022). Color-concept associations reveal an abstract conceptual space. Poster presented at the 22nd Annual Meeting of the Vision Sciences Society.
- 2021 **Mukherjee, K.**, Rogers, T.T., Lessard, L., Gleicher, M., & Schloss, K. B. (2021). Mapping a low-dimensional space of color-concept associations. Poster presented at the 21st Annual Meeting of the Vision Sciences Society. *\*Vision Sciences Society Travel Award*
- 2021 **Mukherjee, K.**, Yin, B., Sherman B. E., Lessard, L. & Schloss, K. B. (2021). Context matters: Semantic discriminability theory for perceptual encoding systems. Talk presented at the 62nd Annual Meeting of the Psychonomic Society.
- 2021 **Mukherjee, K.**, Yin, B., Sherman B. E., Lessard, L. & Schloss, K. B. (2021). Context matters: Semantic discriminability theory for perceptual encoding systems. Talk presented at IEEE VIS 2021.
- 2020 **Mukherjee, K.**, & Rogers, T. T. (2020). How does task structure shape representations in deep neural networks? Poster presented at the 2nd NeurIPS Workshop on Shared Visual Representations in Human and Machine Intelligence.
- 2020 **Mukherjee, K.**, & Rogers, T. T. (2020). Finding meaning in simple sketches: How do humans and deep networks compare? Poster presented at the 20th Annual Meeting of the Vision Sciences Society.
- 2019 **Mukherjee, K.**, Hawkins, R. D., & Fan, J. E. (2019). Communicating semantic part information in drawings. Poster presented at the 41st Annual Meeting of the Cognitive Science Society.

## Invited Talks & Seminars

- 2025 Cognitive abstractions for visual communication, *McPherson Eye Research Institute Seminar*
- 2025 What makes different graphs effective for answering different questions? *Nightingale Labs, Stanford University.*
- 2025 Perceptual semantics for visual and haptic communication. *Wisconsin Institute for Discovery Seminar Series.*
- 2024 EncQA: Evaluating visual encoding understanding for visualizations in large vision-language models, *Apple Human Centered Machine Intelligence Sync.*
- 2023 Using drawings to understand human semantic cognition, MRC Cognition and Brain Sciences Unit, *University of Cambridge.*
- 2023 THINGS-drawings: A large-scale dataset containing human sketches of 1,854 object concepts, Cognitive Brown Bag, *University of Chicago.*
- 2023 THINGS-drawings: A large-scale dataset containing human sketches of 1,854 object concepts, Cognitive Tools Lab, *UC San Diego.*
- 2023 Evaluating machine comprehension of sketch meaning at different levels of abstraction, Stanford NeuroAI Lab, *Stanford University.*
- 2023 Tutorial on matrix completion techniques for the behavioral sciences, *AI and Society Seminar, UW-Madison.*
- 2023 Using drawings and deep neural networks to characterize the building blocks of human visual similarity, *Wisconsin Institute for Discovery Seminar Series.*
- 2022 Using line drawings to understand what deep learning models see, *McPherson Eye Research Institute Seminar*

## Teaching

GRADUATE TEACHING ASSISTANT, UNIVERSITY OF WISCONSIN-MADISON

- 2022 PSYCH 454, *Behavioral Neuroscience*
- 2021 PSYCH 210, *Statistics for Psychology*
- 2020 PSYCH 414, *Cognitive Psychology*

UNDERGRADUATE TEACHING ASSISTANT, VASSAR COLLEGE

- 2017 COGS 211, *Perception and Action*

## Advising

UNDERGRADUATE STUDENTS

- 2025 - Avery Yue (Stanford)
- 2025 - David Tomz (Stanford)
- 2025 - Zach Studdiford (UW-Madison)
- 2024 - 2025 Rainy Jin (UW-Madison)
- 2024 - 2025 Halle Braun (UW-Madison)
- 2024 - 2025 Ria Pattekar (UW-Madison)
- 2024 - 2025 Ankit Mohapatra (UW-Madison)

|             |                                 |
|-------------|---------------------------------|
| 2023 - 2024 | Nancy Davis (UW-Madison)        |
| 2022 - 2023 | Jonah Manaligold (UW-Madison)   |
| 2022 - 2023 | Janani Sundar (UW-Madison)      |
| 2022 - 2023 | Rio Aguina-Kang (UCSD)          |
| 2022        | Lisa Padua (Albany State)       |
| 2020 - 2021 | Brianne E. Sherman (UW-Madison) |

## Professional Service

### WORKSHOP ORGANIZATION

|      |                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | COGGRAPH: Building bridges between cognitive science and computer graphics, <i>46th Annual Meeting of the Cognitive Science Society</i> |
| 2022 | Images2Symbols: Drawing as a Window into the Mind, <i>44th Annual Meeting of the Cognitive Science Society</i>                          |

### PROGRAM COMMITTEES

|      |                                                                          |
|------|--------------------------------------------------------------------------|
| 2025 | VIS×AI: Workshop on Visualization for AI Explainability, <b>IEEE VIS</b> |
|------|--------------------------------------------------------------------------|

### AD HOC REVIEWING

#### **Journals & Books**

Nature, Cognition, Open Mind, Communications Biology, Nature Reviews Psychology, Visualization Psychology

#### **Conference Proceedings and Workshops**

NeurIPS Workshop on Shared Visual Representations in Humans and Machines (SVRHM), Conference on Computational Cognitive Neuroscience (CCN), IEEE Vis (VIS), ACM Conference on Human Factors in Computing Systems (CHI)

### DEPARTMENTAL SERVICE

|           |                                                                  |
|-----------|------------------------------------------------------------------|
| 2020-2022 | University of Wisconsin-Madison Psychology Colloquium Committee  |
| 2017-2019 | Vassar College Cognitive Science Majors' Committee, <i>Chair</i> |
| 2016-2017 | Vassar College Student Association Finance Committee             |

### AFFILIATIONS

|           |                           |
|-----------|---------------------------|
| 2019-     | Cognitive Science Society |
| 2020-     | Vision Sciences Society   |
| 2021-2022 | Psychonomics Society      |

## References

**Dr. Timothy T. Rogers** | University of Wisconsin-Madison, Department of Psychology  
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